After some months of on-and-off mathematical struggle, they proved that there indeed *is* a fully self-consistent billiard ball trajectory that begins with Polchinski's initial data and satisfies all the laws of physics that govern classical billiard balls. In fact, there are *two* such trajectories. They are shown in Figure 14.10. I shall describe each of these trajectories in turn, from the viewpoint of the ball itself.

On trajectory (a) (left half of Figure 14.10), the ball, young, clean, and pristine, starts out at time t = 3:00 P.M., moving along precisely the same route as in Polchinski's paradox (Figure 14.9), a route taking it toward the wormhole's right mouth. A half hour later, at t = 3:30, the young, pristine ball gets hit on its *left, rear side*, by an older-looking, cracked ball (which will turn out to be its older self). The collision is gentle enough to deflect the young ball only slightly from its original course, but hard enough to crack it. The young ball, now cracked, continues onward along its slightly altered trajectory and enters the wormhole mouth at t = 3:45, travels backward in time by 30 minutes, and exits from the other mouth at t = 3:15. Because its trajectory has been altered slightly by comparison with Polchinski's paradoxical trajectory (Figure 14.9), the ball, now old and cracked, hits its younger self a gentle, glancing blow on the left, rear side at t = 3:30, instead of the vigorous, highly deflecting blow of Figure 14.9. The evolution thereby is made fully self-consistent.

Trajectory (b), the right half of Figure 14.10, is the same as (a), except that the geometry of the collision is slightly different, and correspondingly the trajectory between collisions is slightly different. In particular, the old, cracked ball emerges from the left mouth on a different trajectory than in (a), a trajectory that takes it in front of the young, pristine ball (instead of behind it), and produces a glancing blow on the young ball's *front, right side* (instead of left rear side).

Echeverria and Klinkhammer showed that both trajectories, (a) and (b), satisfy all the physical laws that govern classical billiard balls, so both are possible candidates to occur in the real Universe (*if* the real Universe can have wormhole-based time machines).

This is most disquieting. Such a situation can never occur in a universe without time machines. Without time machines, each set of initial conditions for a billiard ball gives rise to one and only one trajectory that satisfies all the classical laws of physics. There is a unique prediction for